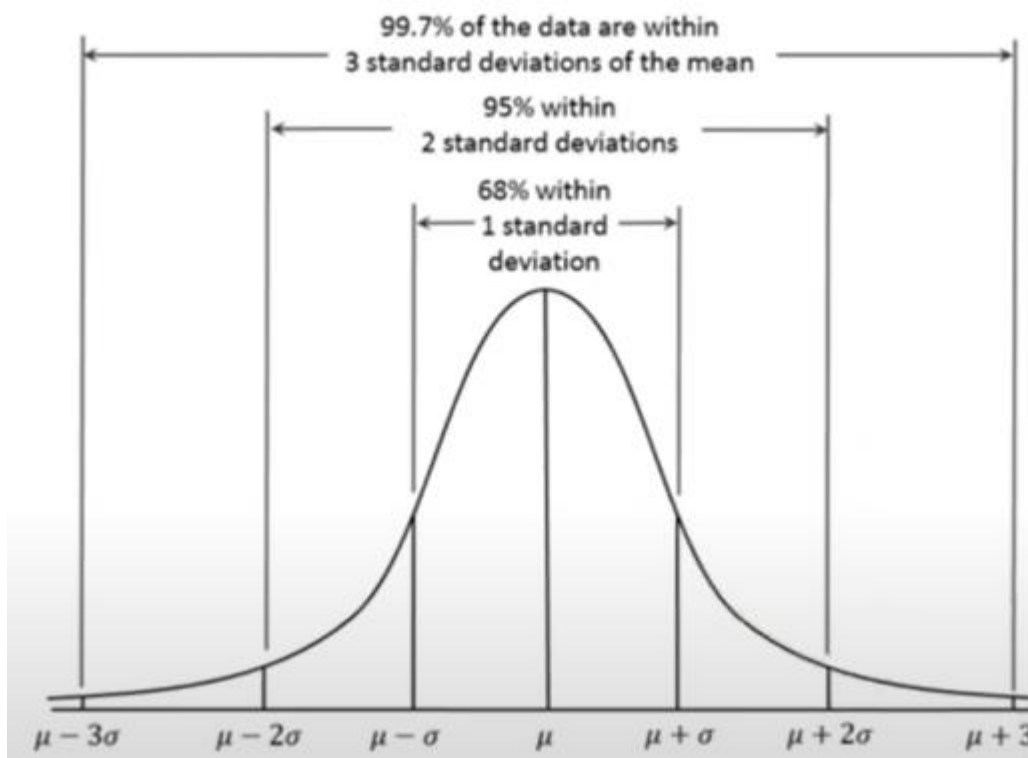


Machine Learning (CAI-262)

Objectives:

- Probability Density Function (PDF)
- Variance- covariance Matrix
- Gaussian Mixture Model
- Expectation Maximization Algorithm

Probability Density Function:



This is the formula for Gaussian distribution also called probability density function (pdf) in which for each x point we calculate y . It is a function of a continuous random variable whose integral across an interval gives the probability that the value of the variable lies within the same interval.

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x - \mu)^2}{2\sigma^2}}$$

Where

μ = Mean

σ = Standard Deviation

$\pi \approx 3.14159 \dots$

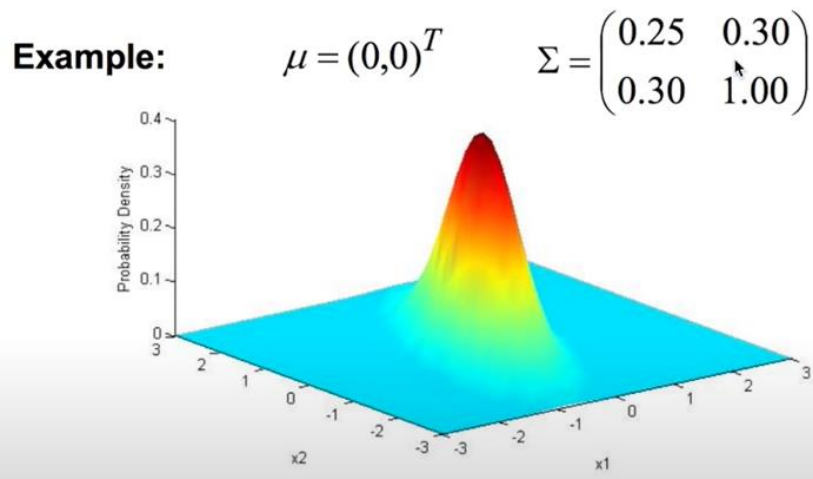
$e \approx 2.71828 \dots$

What is Gaussian Mixture Model?

It is a probability distribution that consists of multiple probability distribution. For d dimensions, the Gaussian distribution of a vector $\mathbf{x}=[x_1, x_2, x_3, \dots, x_d]$ is defined as

$$N(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2} \sqrt{|\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

Where $\boldsymbol{\mu}$ is the mean and $\boldsymbol{\Sigma}$ is the covariance matrix of Gaussian.



Variance-Covariance Matrix

Variance and covariance are often displayed together in a variance-covariance matrix, (also called covariance matrix). The variance appears along diagonal and covariances appear in the off diagonal elements, as show below

$$\mathbf{V} = \begin{bmatrix} \sum x_1^2 / N & \sum x_1 x_2 / N & \dots & \sum x_1 x_c / N \\ \sum x_2 x_1 / N & \sum x_2^2 / N & \dots & \sum x_2 x_c / N \\ \dots & \dots & \dots & \dots \\ \sum x_c x_1 / N & \sum x_c x_2 / N & \dots & \sum x_c^2 / N \end{bmatrix}$$

Where,

$\mathbf{V}$ is a $c \times c$ variance-covariance matrix

N is the number of scores in each of the c data sets

x_i is a deviation score from the i th data set

$\sum x_i^2 / N$ is the variance of elements from the i th data set

$\sum x_i x_j / N$ is the covariance for elements from the i th and j th data sets

The method which uses multiple functions to represent certain data is known as the mixture model. The probability given in a mixture of K Gaussians is as follows.

$$p(x) = \sum_{i=1}^K \pi_i N(x|\mu_i, \Sigma_i)$$

If you consider 2 Gaussian then above formula would look like

$$p(x) = \pi_0 N(x|\mu_0, \Sigma_0) + \pi_1 N(x|\mu_1, \Sigma_1)$$

There are several learnable parameters before a mixture of Gaussians could fit the given data or clusters could be modelled using Gaussians.

- 1) K: How many Gaussians should be used to represent the data.
- 2) π_k mixture parameter for each Gaussian.
- 3) where each Gaussian should be centered μ_k
- 4) what should be spread of each Gaussian σ_k

To learn these parameters we will use an algorithm called Expectation Maximization Algorithm.

Expectation Maximization Algorithm:

Before exploring the EM algorithm, let's understand the central concept behind the EM algorithm called a **latent variable/ hidden variable/ random variable**.

The most important concept to understand EM is called latent variable, which is the ratio of the probability of a certain point belonging to kth Gaussian by the sum of the total probabilities of points belonging to all Gaussians as shown below

$$\gamma_k^i = \frac{P_k(x_i|\mu_k, \Sigma_k)}{\sum_{j=1}^K P_j(x_i|\mu_j, \Sigma_j)}$$

Where latent variable γ_i^k is the probability that point x_i was generated from k^{th} Gaussian. It is important to note that the sum of all mixture weights is 1 that is $\sum_{i=1}^K \pi_i = 1$.

$\pi_i \geq 0$.

To understand the concept Lets assume N1 and N2 are two Gaussians then Latent variables determine the probability of x_i belonging to N1 or N2. Let the probability of P1 that the x_i was generated from N1, and the probability of P2 that the x_i was generated from N2. The generalized equation can be expressed for two Gaussian variables as follows:

$$\gamma_k^i = \frac{P_k(x_i | \mu_k, \Sigma_k)}{P_1(x_i | \mu_1, \Sigma_1) + P_2(x_i | \mu_2, \Sigma_2)}$$

Using the above equation the probabilities could be defined as

$$\gamma_1^i = \frac{P_1}{P_1 + P_2}$$

$$\gamma_2^i = \frac{P_2}{P_1 + P_2}$$

The EM algorithm involves two steps in each iteration: the **”Expectation”** and the **”Maximization”** steps. In the Expectation step, the latent variables γ are estimated. In the Maximization step, the parameters μ , Σ , and π of the model are updated to maximize the expected value of the complete data log-likelihood, based on the γ obtained in the Expectation step. The EM algorithm iterates between these two steps until convergence is achieved. At convergence, the algorithm returns the estimated parameters of the model. The first step of EM is to assume random values for π_k , μ_k , and Σ_k .

After calculating the value of γ_k for each x we use them to correct the initial random assumption of μ , Σ , and π using following formulas.

$$\hat{\mu}_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_k^i x_i$$

$$\hat{\Sigma}_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_k^i (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T$$

$$\hat{\pi}_k = \frac{1}{N} \sum_{i=1}^N \gamma_k^i$$

In M step we determine by applying log maximum likelihood ratio on the mixture model as shown in equation below.

$$\ln p(x|\mu, \Sigma, \pi) = \sum_{i=1}^N \ln \left\{ \sum_{k=1}^K \pi_k N(x_k|\mu_k, \Sigma_k) \right\}$$

Here is the complete algorithm

Algorithm 3 EM algorithm

Require: $X = \{x_1, x_2, \dots, x_N\}$

Ensure: $\forall_k : \{\mu_k, \Sigma_k, \pi_k\}$

[1-] Initialize Mean value μ_k , Mean value Σ_k , mixture weights π_k with random values.

while Covered **do**

[2-] Compute $\gamma_k^i = \frac{\pi_k N(x|\mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j N(x|\mu_j, \Sigma_j)}$ ▷ The E Step

[3-] Mean $\hat{\mu}_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_k^i x_i$, Variances $\hat{\Sigma}_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_k^i (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T$, Weights $\hat{\pi}_k = \frac{1}{N} \sum_{i=1}^N \gamma_k^i$ ▷ The M Step

[4-] Compute $\ln p(x|\mu, \Sigma, \pi)_{OLD} = \ln p(x|\mu, \Sigma, \pi)_{NEW}$ Where,
 $\ln p(x|\mu, \Sigma, \pi)_{NEW} = \sum_{i=1}^N \ln \left\{ \sum_{k=1}^K \pi_k N(x_k|\mu_k, \Sigma_k) \right\}$ ▷ The M Step

end while

Example:

Consider a set of points $x = \{-2, -1, 0, 1, 2\}$ and Two Gaussians i.e. $K = 2$. Apply EM Algorithm.

Step-1: Assume random $\mu_1 = -1, \mu_2 = 1, \sigma_1 = \sigma_2 = 1$ and $\pi_1 = \pi_2 = 0.5$.

Step-2: Calculate latent variable.

$$\gamma_i^k = \frac{\pi_k N(x|\mu_k, \sigma_k)}{\sum_{j=1}^K \pi_j N(x|\mu_j, \sigma_j)}$$

Derive the general value of for Gaussian $\mu_1 = -1$ and $\sigma_1 = 1$.

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}} \implies f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x+1)^2}$$

$$\implies \gamma_i^1 = \frac{\pi \times \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x+1)^2}}{\pi \times \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x+1)^2} + \pi \times \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-1)^2}}$$

$$\implies \gamma_i^1 = \frac{\cancel{\pi} \times \frac{1}{\cancel{\sqrt{2\pi}}} e^{-\frac{1}{2}(x+1)^2}}{\cancel{\pi} \times \frac{1}{\cancel{\sqrt{2\pi}}} \{e^{-\frac{1}{2}(x+1)^2} + e^{-\frac{1}{2}(x-1)^2}\}}$$

$$\implies \gamma_i^1 = \frac{e^{-\frac{1}{2}(x+1)^2}}{e^{-\frac{1}{2}(x+1)^2} + e^{-\frac{1}{2}(x-1)^2}}$$

For -2:

$$\gamma_{-2}^1 = \frac{e^{-\frac{1}{2}(-2+1)^2}}{e^{-\frac{1}{2}(-2+1)^2} + e^{-\frac{1}{2}(-2-1)^2}} \Rightarrow 0.98 \quad \gamma_{-2}^2 = 1 - \gamma_{-2}^1 \Rightarrow 0.02$$

For -1:

$$\gamma_{-1}^1 = \frac{e^{-\frac{1}{2}(-1+1)^2}}{e^{-\frac{1}{2}(-1+1)^2} + e^{-\frac{1}{2}(-1-1)^2}} \Rightarrow 0.76 \quad \gamma_{-1}^2 = 1 - \gamma_{-1}^1 \Rightarrow 0.24$$

For 0:

$$\gamma_0^1 = \frac{e^{-\frac{1}{2}(0+1)^2}}{e^{-\frac{1}{2}(0+1)^2} + e^{-\frac{1}{2}(0-1)^2}} \Rightarrow 0.5 \quad \gamma_0^2 = 1 - \gamma_0^1 \Rightarrow 0.5$$

For 1:

$$\gamma_1^1 = \frac{e^{-\frac{1}{2}(1+1)^2}}{e^{-\frac{1}{2}(1+1)^2} + e^{-\frac{1}{2}(1-1)^2}} \Rightarrow 0.11 \quad \gamma_1^2 = 1 - \gamma_1^1 \Rightarrow 0.89$$

For 2:

$$\gamma_2^1 = \frac{e^{-\frac{1}{2}(2+1)^2}}{e^{-\frac{1}{2}(2+1)^2} + e^{-\frac{1}{2}(2-1)^2}} \Rightarrow 0.07 \quad \gamma_2^2 = 1 - \gamma_2^1 \Rightarrow 0.93$$

Step-3: Calculate New μ_j , σ_j and π_j .

$$\hat{\mu}_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_k^i x_i$$

$$\hat{\mu}_1 = \frac{1}{5}[(-2 \times 0.98) + (-1 \times 0.76) + (0 \times 0.5) + (1 \times 0.11) + (2 \times 0.07)]$$

$$\Rightarrow \hat{\mu}_1 = \frac{1}{5}[-1.96 - 0.76 + 0 + 0.11 + 0.14] \quad \Rightarrow \hat{\mu}_1 = -0.49$$

$$\hat{\mu}_2 = \frac{1}{5}[(-2 \times 0.02) + (-1 \times 0.24) + (0 \times 0.5) + (1 \times 0.89) + (2 \times 0.93)]$$

$$\Rightarrow \hat{\mu}_2 = \frac{1}{5}[-0.4 - 0.24 + 0 + 0.89 + 1.86] \quad \Rightarrow \hat{\mu}_2 = 1.02$$

$$\hat{\Sigma}_k = \frac{1}{N_k} \sum_{i=1}^N \gamma_k^i (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T$$

$$\hat{\sigma}_1 = \frac{1}{5}[0.98 \times (-2 + 0.49)^2 + 0.76 \times (-1 + 0.49)^2 + 0.5 \times (0 + 0.49)^2 + 0.11 \times (1 + 0.49)^2 + 0.07 \times (2 + 0.49)^2]$$

$$\Rightarrow \hat{\sigma}_1 = \frac{1}{5}[2.23 + 0.19 + 0.12 + 0.24 + 0.43] \Rightarrow 0.64$$

$$\hat{\sigma}_2 = \frac{1}{5}[0.02 \times (-2 - 1.02)^2 + 0.24 \times (-1 - 1.02)^2 + 0.5 \times (0 - 1.02)^2 + 0.89 \times (1 - 1.02)^2 + 0.93 \times (2 - 1.02)^2]$$

$$\hat{\sigma}_2 = \frac{1}{5} [0.02 \times (-2 - 1.02)^2 + 0.24 \times (-1 - 1.02)^2 + 0.5 \times (0 - 1.02)^2 + 0.89 \times (1 - 1.02)^2 + 0.93 \times (2 - 1.02)^2]$$

$$\Rightarrow \hat{\sigma}_2 = \frac{1}{5} [0.18 + 0.97 + 0.52 + 0.0003 + 0.89] \Rightarrow 0.51$$

$$\Rightarrow \hat{\pi}_1 = \frac{1}{5} [0.98 + 0.76 + 0.5 + 0.11 + 0.07] \Rightarrow 0.48$$

$$\Rightarrow \hat{\sigma}_2 = \frac{1}{5} [0.02 + 0.24 + 0.5 + 0.89 + 0.93] \Rightarrow 0.51$$

$$\hat{\pi}_k = \frac{1}{N} \sum_{i=1}^N \gamma_k^i$$

$$\Rightarrow \hat{\pi}_1 = \frac{1}{5} [0.98 + 0.76 + 0.5 + 0.11 + 0.07] \Rightarrow 0.48$$

$$\Rightarrow \hat{\pi}_2 = \frac{1}{5} [0.02 + 0.24 + 0.5 + 0.89 + 0.93] \Rightarrow 0.51$$